

Solutions to Axler's *Linear Algebra Done Right*, 4th Edition

by: blanketism

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7D Isometries, Unitary Operators, and Matrix Factorization

Exercise 7D1

Suppose $\dim V \geq 2$ and $S \in \mathcal{L}(V, W)$. Prove that S is an isometry if and only if Se_1, Se_2 is an orthonormal list in W for every orthonormal list e_1, e_2 of length two in V .

Solution.

$\Rightarrow$ Let S be an isometry, and let e_1, e_2 be an orthonormal list of length two in V . By 7.49(c), $\langle Se_i, Se_j \rangle = \langle e_i, e_j \rangle$ for all $i, j \in \{1, 2\}$, so Se_1, Se_2 is an orthonormal list in W .

$\Leftarrow$ Let $u \in V$ with $\|u\| = 1$. By 6.36 and $\dim V \geq 2$, extend u to an orthonormal list u, v of length two in V . By hypothesis, Su, Sv is an orthonormal list in W ; in particular, $\|Su\| = 1$. Now let $x \in V$. If $x = 0$, then $\|Sx\| = 0 = \|x\|$. Otherwise, $\frac{x}{\|x\|}$ has norm 1. Then

$$\|Sx\| = \left\| S \left(\frac{x}{\|x\|} \right) \right\| \|x\| = \|x\|.$$

Hence, S preserves norms so that S is an isometry. ■

Exercise 7D2

Suppose $T \in \mathcal{L}(V, W)$ and $T \neq 0$. Prove that T is a scalar multiple of an isometry if and only if T preserves orthogonality.

Axler Note. The phrase “ T preserves orthogonality” means that $\langle Tu, Tv \rangle = 0$ for all $u, v \in V$ such that $\langle u, v \rangle = 0$.

Solution.

$\Rightarrow$ Let $T = \lambda S$, where $\lambda \in \mathbb{F}$ and $S \in \mathcal{L}(V, W)$ is an isometry. Then for any $u, v \in V$ such that $\langle u, v \rangle = 0$, we have

$$\langle Tu, Tv \rangle = \langle \lambda Su, \lambda Sv \rangle = \lambda \bar{\lambda} \langle Su, Sv \rangle = |\lambda|^2 \langle Su, Sv \rangle = |\lambda|^2 \langle u, v \rangle = 0$$

by 7.49(c). Hence, T preserves orthogonality.

$\Leftarrow$ Since $T \neq 0$, $V \neq \{0\}$. Let $e_1, \dots, e_n$ be an orthonormal basis of V , where $n = \dim V \geq 1$. For $j, k \in \{1, \dots, n\}$ with $j \neq k$, we have $\langle e_j, e_k \rangle = 0$, so by assumption $\langle Te_j, Te_k \rangle = 0$. Since $\|e_j\| = \|e_k\|$ and $\langle e_k, e_j \rangle = \langle e_j, e_k \rangle = 0$, it follows that $\langle e_j + e_k, e_j - e_k \rangle = 0$. Therefore,

$$0 = \langle T(e_j + e_k), T(e_j - e_k) \rangle = \langle Te_j + Te_k, Te_j - Te_k \rangle = \langle Te_j, Te_j \rangle - \langle Te_k, Te_k \rangle = \|Te_j\|^2 - \|Te_k\|^2,$$

where $\langle Te_j, Te_k \rangle = \langle Te_k, Te_j \rangle = 0$ for $j \neq k$. Hence, $\|Te_j\| = \|Te_k\|$ for all $j, k \in \{1, \dots, n\}$. Let $\lambda = \|Te_1\|$. Then $\|Te_j\| = \lambda$ for all $j = 1, \dots, n$. If $\lambda = 0$, then $Te_j = 0$ for all j , so $T = 0$ on a basis. This implies $T = 0$, which contradicts the assumption that $T \neq 0$. Hence, $\lambda > 0$. Define $S = \frac{1}{\lambda}T$. Then $\|Se_j\| = 1$ for all $j = 1, \dots, n$. Moreover,

$$\langle Se_j, Se_k \rangle = \left\langle \frac{1}{\lambda}Te_j, \frac{1}{\lambda}Te_k \right\rangle = \frac{1}{\lambda^2} \langle Te_j, Te_k \rangle = 0$$

for $j \neq k$. Hence, $Se_1, \dots, Se_n$ is an orthonormal list in W , and by 7.49(d), S is an isometry. Therefore, $T = \lambda S$ is a scalar multiple of an isometry. ■

Exercise 7D3

- (a) Show that the product of two unitary operators on V is a unitary operator.
 (b) Show that the inverse of a unitary operator on V is a unitary operator.

Axler Note. This exercise shows that the set of unitary operators on V is a group, where the group operation is the usual product of two operators.

Solution.

- (a) Let T_1 and T_2 be unitary operators on V . By 7.5(d) and 7.53(b), we have

$$(T_1 T_2)^*(T_1 T_2) = T_2^* T_1^* T_1 T_2 = T_2^* T_2 = I, \quad (T_1 T_2)(T_1 T_2)^* = T_1 T_2 T_2^* T_1^* = T_1 I T_1^* = T_1 T_1^* = I.$$

Hence, $T_1 T_2$ is a unitary operator by 7.53(b).

- (b) Let T be a unitary operator on V . By 7.53(c), $T^{-1} = T^*$, and T^* is unitary by 7.53(f). ■

Exercise 7D4

Suppose $\mathbb{F} = \mathbb{C}$ and $A, B \in \mathcal{L}(V)$ are self-adjoint. Show that $A + iB$ is unitary if and only if $AB = BA$ and $A^2 + B^2 = I$.

Solution. Let $T = A + iB$. By 7.5(a), (b), and the fact that A and B are self-adjoint, we have $T^* = A - iB$. Therefore,

$$TT^* = A^2 + B^2 + i(BA - AB), \quad T^*T = A^2 + B^2 + i(AB - BA).$$

⇒ Suppose T is unitary. By 7.53(b), $TT^* = T^*T = I$. Then

$$TT^* - T^*T = i(BA - AB) - i(AB - BA) = 2i(BA - AB) = 0,$$

so $AB = BA$. Moreover, $TT^* = I$ implies $A^2 + B^2 + i(BA - AB) = I$, and since $AB = BA$, this simplifies to $A^2 + B^2 = I$.

⇐ Suppose $AB = BA$ and $A^2 + B^2 = I$. Substituting these into either TT^* or T^*T , we get

$$TT^* = T^*T = A^2 + B^2 = I.$$

Hence, T is unitary by 7.53(b). ■

Exercise 7D5

Suppose $S \in \mathcal{L}(V)$. Prove that the following are equivalent.

- (a) S is a self-adjoint unitary operator.
 (b) $S = 2P - I$ for some orthogonal projection P on V .
 (c) There exists a subspace U of V such that $Su = u$ for every $u \in U$ and $Sw = -w$ for every $w \in U^\perp$.

Solution.

(a) ⇒ (b) Suppose S is a self-adjoint unitary operator. Then $S = S^*$. By 7.53(b), $S^*S = SS^* = I$, so $S^2 = I$. Let $P = \frac{1}{2}(S + I)$. By 7.5(a), (b), and (e), P is self-adjoint with

$$P^2 = \left(\frac{1}{2}(S + I)\right)^2 = \frac{1}{4}(S^2 + 2S + I) = \frac{1}{4}(I + 2S + I) = \frac{1}{2}(S + I) = P.$$

Since P is self-adjoint and $P^2 = P$, Exercise 7A20 gives that P is an orthogonal projection. Then $S = 2P - I$.

(b) $\Rightarrow$ (c) Suppose $S = 2P - I$ for some orthogonal projection P onto a subspace U of V . By 6.57, $Pu = u$ for $u \in U$, and $Pw = 0$ for $w \in U^\perp$. Hence,

$$Su = (2P - I)u = 2Pu - u = 2u - u = u, \quad Sw = (2P - I)w = 2Pw - w = 0 - w = -w$$

for $u \in U$ and $w \in U^\perp$.

(c) $\Rightarrow$ (a) Suppose there exists a subspace U of V such that $Su = u$ for every $u \in U$ and $Sw = -w$ for every $w \in U^\perp$. Since V is finite-dimensional, so is U . Let P be the orthogonal projection onto U . For any $v \in V$, write $v = u + w$ with $u \in U$ and $w \in U^\perp$ by 6.49. By 6.57(b) and (c),

$$(2P - I)v = (2P - I)(u + w) = 2P(u + w) - (u + w) = 2u - (u + w) = u - w = Su + Sw = Sv.$$

Hence, $S = 2P - I$. By Exercise 7A20, P is self-adjoint. It follows that $S^* = 2P^* - I = 2P - I = S$ by 7.5. By 6.57, $P^2 = P$, so

$$S^2 = (2P - I)^2 = 4P^2 - 4P + I = 4P - 4P + I = I$$

Therefore, $SS^* = S^*S = S^2 = I$, and S is a self-adjoint unitary operator by 7.53(b). ■

Exercise 7D6

Suppose T_1, T_2 are both normal operators on $\mathbb{F}^3$ with 2, 5, 7 as eigenvalues. Prove that there exists a unitary operator $S \in \mathcal{L}(\mathbb{F}^3)$ such that $T_1 = S^*T_2S$.

Solution. Let e_2, e_5, e_7 and f_2, f_5, f_7 be unit eigenvectors of T_1 and T_2 corresponding to the eigenvalues 2, 5, 7, respectively. By 7.22, each list is an orthonormal list in $\mathbb{F}^3$. By 6.28, each list is an orthonormal basis of $\mathbb{F}^3$. By 3.4, there exists a linear operator $S \in \mathcal{L}(\mathbb{F}^3)$ such that $Se_2 = f_2$, $Se_5 = f_5$, and $Se_7 = f_7$. By 7.53(d), S is unitary, and $S^* = S^{-1}$ by 7.53(c). For each $k \in \{2, 5, 7\}$, we have $T_1e_k = ke_k$ and $T_2f_k = kf_k$, so

$$S^*T_2Se_k = S^*T_2f_k = S^*(kf_k) = kS^*f_k = kS^{-1}f_k = ke_k = T_1e_k.$$

Since T_1 and S^*T_2S agree on an orthonormal basis of $\mathbb{F}^3$, we have $T_1 = S^*T_2S$ by 3.4. ■

Exercise 7D7

Give an example of two self-adjoint operators $T_1, T_2 \in \mathcal{L}(\mathbb{F}^4)$ such that the eigenvalues of both operators are 2, 5, 7, but there does not exist a unitary operator $S \in \mathcal{L}(\mathbb{F}^4)$ such that $T_1 = S^*T_2S$. Be sure to explain why there is no unitary operator with the required property.

Solution. Let e_1, e_2, e_3, e_4 be the standard basis of $\mathbb{F}^4$, which is an orthonormal basis. Consider the operators $T_1, T_2 \in \mathcal{L}(\mathbb{F}^4)$ whose matrix representations with respect to e_1, e_2, e_3, e_4 are diagonal with diagonal entries 2, 2, 5, 7 and 2, 5, 5, 7, respectively. By 7.9, both operators are self-adjoint since real diagonal matrices equal their conjugate transposes. By 5.41, the eigenvalues of T_1 and T_2 are exactly 2, 5, 7. Note that $E(2, T_1) = \text{null}(T_1 - 2I) = \text{span}(e_1, e_2)$ and $E(2, T_2) = \text{null}(T_2 - 2I) = \text{span}(e_1)$.

Now suppose $S \in \mathcal{L}(\mathbb{F}^4)$ is a unitary operator such that $T_1 = S^*T_2S$. By 7.53(c), $S^* = S^{-1}$. Then $ST_1 = S(S^*T_2S) = (SS^{-1})T_2S = T_2S$. For any $v \in E(2, T_1)$, we have $T_1v = 2v$. Thus,

$$T_2(Sv) = S(T_1v) = S(2v) = 2Sv,$$

which shows that S maps $E(2, T_1)$ into $E(2, T_2)$. Moreover, $\dim E(2, T_1) = 2$ and $\dim E(2, T_2) = 1$. By 3.22, $S|_{E(2, T_1)}$ cannot be injective, and therefore S cannot be injective. Since a unitary operator must be invertible by 7.53(c), this is a contradiction. Hence, no such unitary operator exists. ■

Exercise 7D8

Prove or give a counterexample: If $S \in \mathcal{L}(V)$ and there exists an orthonormal basis $e_1, \dots, e_n$ of V such that $\|Se_k\| = 1$ for each e_k , then S is a unitary operator.

Solution. The statement is false in general. Consider $V = \mathbb{F}^2$ with the standard orthonormal basis e_1, e_2 . Define $S \in \mathcal{L}(V)$ by $Se_1 = Se_2 = e_1$. Then $\|Se_1\| = \|Se_2\| = 1$, but S is not unitary since $S(e_1 - e_2) = 0$, which shows that S is not injective and therefore not invertible. ■

Exercise 7D9

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Suppose every eigenvalue of T has absolute value 1 and $\|Tv\| \leq \|v\|$ for every $v \in V$. Prove that T is a unitary operator.

Solution. If $V = \{0\}$, then T is trivially unitary. Otherwise, $V \neq \{0\}$, so by Schur's theorem, there exists an orthonormal basis $e_1, \dots, e_n$ of V such that $M(T)$ is upper triangular with respect to this basis; call this matrix A . Then for each $k = 1, \dots, n$, $Te_k = \sum_{j=1}^k A_{j,k}e_j$. By 5.41, the diagonal entries $A_{k,k}$ are the eigenvalues of T , so $|A_{k,k}| = 1$ for each $k = 1, \dots, n$. By 6.24 and the hypothesis,

$$1 \geq \|Te_k\|^2 = \sum_{j=1}^k |A_{j,k}|^2 = |A_{k,k}|^2 + \sum_{j=1}^{k-1} |A_{j,k}|^2 = 1 + \sum_{j=1}^{k-1} |A_{j,k}|^2,$$

which implies $0 \geq \sum_{j=1}^{k-1} |A_{j,k}|^2$. Then $A_{j,k} = 0$ for all $j < k$, so A is diagonal, and $Te_k = A_{k,k}e_k$ for each $k = 1, \dots, n$. It follows that $e_1, \dots, e_n$ is an orthonormal basis of eigenvectors of T with eigenvalues of absolute value 1. By 7.55, T is unitary. ■

Exercise 7D10

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$ is a self-adjoint operator such that $\|Tv\| \leq \|v\|$ for all $v \in V$.

- Show that $I - T^2$ is a positive operator.
- Show that $T + i\sqrt{I - T^2}$ is a unitary operator.

Solution.

- By 7.5, $I - T^2$ is self-adjoint. Then for any $v \in V$,

$$\langle (I - T^2)v, v \rangle = \langle v, v \rangle - \langle T^2v, v \rangle = \langle v, v \rangle - \langle Tv, T^*v \rangle = \langle v, v \rangle - \langle Tv, Tv \rangle = \|v\|^2 - \|Tv\|^2 \geq 0,$$

where the final inequality follows from $\|Tv\| \leq \|v\|$ for all $v \in V$. Hence, $I - T^2$ is a positive operator.

- Let $B = \sqrt{I - T^2}$. By part (a) and 7.39, B is a positive self-adjoint operator. By Exercise 7C17, there exists a polynomial p such that $B = p(I - T^2)$, where p has real coefficients. Since T commutes

with $I - T^2$, it commutes with B . Moreover,

$$T^2 + B^2 = T^2 + (I - T^2) = I.$$

By Exercise 7D4 $\Leftarrow$ with $A = T$ and B as above, $T + iB$ is a unitary operator. ■

Exercise 7D11

Suppose $S \in \mathcal{L}(V)$. Prove that S is a unitary operator if and only if

$$\{Sv \mid v \in V \text{ and } \|v\| \leq 1\} = \{v \in V \mid \|v\| \leq 1\}.$$

Solution. Consider the set $\mathcal{B} = \{v \in V \mid \|v\| \leq 1\}$. For any $T \in \mathcal{L}(V)$, let $T(\mathcal{B}) = \{Tv \mid v \in \mathcal{B}\}$. Now, if T is such that $T(\mathcal{B}) \subseteq \mathcal{B}$, then $\|Tv\| \leq 1$ for all $v \in \mathcal{B}$. If $v = 0$, then $\|T0\| = 0 = \|0\|$. If $v \neq 0$, then

$$\frac{\|Tv\|}{\|v\|} = \left\| T\left(\frac{v}{\|v\|}\right) \right\| \leq 1.$$

Hence, $\|Tv\| \leq \|v\|$ for all $v \in V$.

$\Rightarrow$ Suppose S is a unitary operator. Then S is also an isometry, so $S(\mathcal{B}) \subseteq \mathcal{B}$. By Exercise 7D3(b), S^{-1} is also unitary. Then for any $u \in \mathcal{B}$, $S^{-1}u \in \mathcal{B}$. Therefore, $u = S(S^{-1}u) \in S(\mathcal{B})$, so $S(\mathcal{B}) = \mathcal{B}$.

$\Leftarrow$ Now suppose $S(\mathcal{B}) = \mathcal{B}$. Let $w \in V$, and write $w = \|w\|w_0$ for some $w_0 \in \mathcal{B}$. Since $\mathcal{B} = S(\mathcal{B})$, there exists $v_0 \in \mathcal{B}$ such that $w_0 = Sv_0$. Hence, $w = \|w\|Sv_0 = S(\|w\|v_0)$, so $w \in \text{range } S$. Therefore, S is surjective, and hence invertible by 3.65. Also, if $u \in \mathcal{B}$, then $u = Sv$ for some $v \in \mathcal{B}$. Hence, $v = S^{-1}u \in \mathcal{B}$, so $S^{-1}(\mathcal{B}) \subseteq \mathcal{B}$. Applying the preliminary result from the beginning of the solution to S and S^{-1} , we get $\|Sx\| \leq \|x\|$ and $\|x\| = \|S^{-1}(Sx)\| \leq \|Sx\|$, so $\|Sx\| = \|x\|$ for all $x \in V$. Therefore, S is an isometry, and since it is also invertible, it is unitary. ■

Exercise 7D12

Prove or give a counterexample: If $S \in \mathcal{L}(V)$ is invertible and $\|S^{-1}v\| = \|Sv\|$ for every $v \in V$, then S is unitary.

Manual Remark. The solution below uses 2 and $\frac{1}{2}$, but replacing them with λ and $\frac{1}{\lambda}$ for any nonzero $\lambda \in \mathbb{F}$ with $|\lambda| \neq 1$ gives a similar counterexample.

Solution. The statement is false. Consider $\mathbb{F}^2$ with the Euclidean inner product, and let $S \in \mathcal{L}(\mathbb{F}^2)$ be defined by $S(x, y) = (2y, \frac{1}{2}x)$. Then

$$S^2(x, y) = S\left(2y, \frac{1}{2}x\right) = \left(2 \cdot \frac{1}{2}x, \frac{1}{2} \cdot 2y\right) = (x, y).$$

Hence, S is invertible with $S^{-1} = S$. Then $\|S^{-1}(x, y)\| = \|S(x, y)\|$ for all $(x, y) \in \mathbb{F}^2$. However, $\|S(0, 1)\| = 2 \neq 1 = \|(0, 1)\|$. Therefore, S is not unitary. ■

Exercise 7D13

Explain why the columns of a square matrix of complex numbers form an orthonormal list in $\mathbb{C}^n$ if and only if the rows of the matrix form an orthonormal list in $\mathbb{C}^n$.

Solution. Let Q be an n -by- n matrix with complex entries. Define $T \in \mathcal{L}(\mathbb{C}^n)$ by $Tv = Qv$ for all $v \in \mathbb{C}^n$. Moreover, let $\mathbb{C}^n$ have the Euclidean inner product. If $e_1, \dots, e_n$ denotes the standard (orthonormal) basis of $\mathbb{C}^n$, then Te_k is the k th column of Q . Hence, $\mathcal{M}(T) = Q$ with respect to $e_1, \dots, e_n$. Then

$$\begin{aligned}
 Q \text{ has orthonormal columns} &\iff Te_1, \dots, Te_n \text{ is an orthonormal list} \\
 &\iff Te_1, \dots, Te_n \text{ is an orthonormal basis by 6.28} \\
 &\iff T \text{ is unitary by 7.53(d)} \\
 &\iff \text{the rows of } \mathcal{M}(T) = Q \text{ form an orthonormal basis by 7.53(e)} \\
 &\iff \text{the rows of } Q \text{ form an orthonormal list by 6.28.} \quad \blacksquare
 \end{aligned}$$

Exercise 7D14

Suppose $v \in V$ with $\|v\| = 1$ and $b \in \mathbb{F}$. Also suppose $\dim V \geq 2$. Prove that there exists a unitary operator $S \in \mathcal{L}(V)$ such that $\langle Sv, v \rangle = b$ if and only if $|b| \leq 1$.

Solution.

$\Rightarrow$ Note that $\|Sv\| = \|v\| = 1$ since S is unitary. By the Cauchy–Schwarz inequality, $|b| = |\langle Sv, v \rangle| \leq \|Sv\| \|v\| = 1$.

$\Leftarrow$ Suppose $|b| \leq 1$. By 6.36, we may extend v to an orthonormal basis $v, u, e_3, \dots, e_n$ of V , which has length $\dim V \geq 2$. Let $\beta = \sqrt{1 - |b|^2}$. By 3.4, there exists an operator $S \in \mathcal{L}(V)$ such that

$$Sv = bv + \beta u, \quad Su = -\beta v + \bar{b}u, \quad Se_k = e_k \text{ for } k \geq 3.$$

Then

$$\|Sv\|^2 = \|bv + \beta u\|^2 = |b|^2\|v\|^2 + |\beta|^2\|u\|^2 = |b|^2 + (1 - |b|^2) = 1,$$

and

$$\|Su\|^2 = \|-\beta v + \bar{b}u\|^2 = |\beta|^2\|v\|^2 + |\bar{b}|^2\|u\|^2 = (1 - |b|^2) + |b|^2 = 1.$$

Moreover, noting that $\bar{\bar{b}} = b$, we have

$$\langle Sv, Su \rangle = \langle bv + \beta u, -\beta v + \bar{b}u \rangle = b(-\beta)\langle v, v \rangle + b^2\langle v, u \rangle - \beta^2\langle u, v \rangle + \beta b\langle u, u \rangle = -b\beta + \beta b = 0.$$

Lastly, $Sv, Su \in \text{span}(v, u)$, which is orthogonal to all e_k for $k \geq 3$. Then the list $Sv, Su, Se_3, \dots, Se_n$ is an orthonormal basis of V by 6.28, so S is unitary by 7.53(d). Finally,

$$\langle Sv, v \rangle = \langle bv + \beta u, v \rangle = b\langle v, v \rangle + \beta\langle u, v \rangle = b. \quad \blacksquare$$

Exercise 7D15

Suppose T is a unitary operator on V such that $T - I$ is invertible.

(a) Prove that $(T + I)(T - I)^{-1}$ is a skew operator (meaning that it equals the negative of its adjoint).

(b) Prove that if $\mathbb{F} = \mathbb{C}$, then $i(T + I)(T - I)^{-1}$ is a self-adjoint operator.

Axler Note. The function $z \mapsto i(z + 1)(z - 1)^{-1}$ maps the unit circle in $\mathbb{C}$ (except for the point 1) to $\mathbb{R}$. Thus (b) illustrates the analogy between the unitary operators and the unit circle in $\mathbb{C}$, along with the analogy between the self-adjoint operators and $\mathbb{R}$.

Solution. Define $A = (T + I)(T - I)^{-1}$. Observe that $T + I = (T - I) + 2I$. Then $A = I + 2(T - I)^{-1}$.

(a) By 7.5, $A^* = I + 2(T^* - I)^{-1}$. By 7.53(c), $T^* = T^{-1}$, so $T^* - I = -T^{-1}(T - I)$, and

$$(T^* - I)^{-1} = -(T - I)^{-1}T = -(T - I)^{-1}((T - I) + I) = -I - (T - I)^{-1}.$$

Therefore, $A^* = -I - 2(T - I)^{-1} = -A$.

(b) Using 7.5(b) and part (a), $(iA)^* = iA^* = (-i)(-A) = iA$. ■

Exercise 7D16

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$ is self-adjoint. Prove that $(T + iI)(T - iI)^{-1}$ is a unitary operator and 1 is not an eigenvalue of this operator.

Solution. Observe that every eigenvalue of T is real by 7.12. Therefore, neither i nor $-i$ is an eigenvalue of T so that $T - iI$ and $T + iI$ are invertible. Define $S = (T + iI)(T - iI)^{-1}$. By 7.5, $(T + iI)^* = T^* - iI = T - iI$ and $((T - iI)^{-1})^* = ((T - iI)^*)^{-1} = (T + iI)^{-1}$. Thus,

$$S^* = (T + iI)^{-1}(T - iI).$$

Because $T - iI$ and $T + iI$ commute,

$$S^*S = (T + iI)^{-1}(T - iI)(T + iI)(T - iI)^{-1} = I,$$

showing that S is an isometry on V by 7.49(b), whence S is unitary.

Now, suppose $Sx = x$ for some $x \in V$. By definition of S , $(T + iI)(T - iI)^{-1}x = x$, so $(T + iI)y = x$ where $y = (T - iI)^{-1}x$. Since $x = (T - iI)y$, we have $(T + iI)y = (T - iI)y$, so $2iy = 0$, and hence $y = 0$. Therefore, $x = (T - iI)y = 0$, showing that 1 is not an eigenvalue of S . ■

Exercise 7D17

Explain why the characterizations of unitary matrices given by 7.57 hold.

Solution. Let Q be an n -by- n matrix. Define $T \in \mathcal{L}(\mathbb{F}^n)$ by $Tv = Qv$ for all $v \in \mathbb{F}^n$, so $\mathcal{M}(T) = Q$ with respect to the standard basis of $\mathbb{F}^n$, which is orthonormal with respect to the Euclidean inner product.

(a) $\Leftrightarrow$ (c) Since $Tv = Qv$ for all $v \in \mathbb{F}^n$, (c) states precisely that T is an isometry. By the equivalence of 7.49(a) and (e), T is an isometry if and only if the columns of $\mathcal{M}(T) = Q$ are orthonormal, which is (a) by definition.

(a) $\Leftrightarrow Q^*Q = I$, (b) $\Leftrightarrow QQ^* = I$ For $j, k \in \{1, \dots, n\}$, consider the j, k th entry of both Q^*Q and QQ^* . By 3.46 and the definition of the conjugate transpose, we have

$$(Q^*Q)_{j,k} = \sum_{\ell=1}^n \overline{Q_{\ell,j}} Q_{\ell,k} = \langle Q_{\cdot,j}, Q_{\cdot,k} \rangle, \quad (QQ^*)_{j,k} = \sum_{\ell=1}^n Q_{j,\ell} \overline{Q_{k,\ell}} = \langle Q_{j,\cdot}, Q_{k,\cdot} \rangle.$$

Note that $Q_{j,\cdot}$ and $Q_{k,\cdot}$ are treated as vectors in $\mathbb{F}^n$. Therefore, $Q^*Q = I$ if and only if the columns of Q are orthonormal, which is (a), and $QQ^* = I$ if and only if the rows of Q are orthonormal, which is (b).

$Q^*Q = I \Leftrightarrow QQ^* = I$ By 7.9 and 3.43, $\mathcal{M}(T^*T) = Q^*Q$ and $\mathcal{M}(TT^*) = QQ^*$. Then $Q^*Q = I$ if and only if $T^*T = I$, and $QQ^* = I$ if and only if $TT^* = I$. By 3.68, $T^*T = I$ if and only if $TT^* = I$. Therefore, $Q^*Q = I$ if and only if $QQ^* = I$, so each condition is equivalent to (d).

This shows that (a) is equivalent to (c) and that (a) is equivalent to (d), which is equivalent to (b). ■

Exercise 7D18

A square matrix A is called **symmetric** if it equals its transpose. Prove that if A is a symmetric matrix with real entries, then there exists a unitary matrix Q with real entries such that Q^*AQ is a diagonal matrix.

Solution. Let A be a symmetric n -by- n matrix with real entries. Define $T \in \mathcal{L}(\mathbb{R}^n)$ by $Tv = Av$ for all $v \in \mathbb{R}^n$, so $\mathcal{M}(T) = A$ with respect to the standard orthonormal basis of $\mathbb{R}^n$. Since A is real and symmetric, $A^* = A^t = A$. Then $\mathcal{M}(T^*) = A^* = A = \mathcal{M}(T)$, so T is self-adjoint by 7.9. By the real spectral theorem, there exists an orthonormal basis $f_1, \dots, f_n$ of $\mathbb{R}^n$ such that $Tf_k = \lambda_k f_k$ for some real numbers $\lambda_1, \dots, \lambda_n$.

Define Q to be the matrix whose columns are the vectors $f_1, \dots, f_n$. Moreover, let D be the diagonal matrix with entries $\lambda_1, \dots, \lambda_n$ on its diagonal. By definition, Q is a matrix with real entries and orthonormal columns, so it is unitary. Moreover, the k th column of AQ is $Af_k = Tf_k = \lambda_k f_k$, which is also the k th column of QD for each $k = 1, \dots, n$. Hence, $AQ = QD$. By 7.57(d), $Q^*Q = I$, so $Q^*AQ = Q^*QD = D$. ■

Exercise 7D19

Suppose n is a positive integer. For this exercise, we adopt the notation that a typical element z of $\mathbb{C}^n$ is denoted by $z = (z_0, z_1, \dots, z_{n-1})$. Define linear functionals $\omega_0, \omega_1, \dots, \omega_{n-1}$ on $\mathbb{C}^n$ by

$$\omega_j(z_0, z_1, \dots, z_{n-1}) = \frac{1}{\sqrt{n}} \sum_{m=0}^{n-1} z_m e^{-2\pi i j m / n}.$$

The **discrete Fourier transform** is the operator $\mathcal{F} : \mathbb{C}^n \rightarrow \mathbb{C}^n$ defined by

$$\mathcal{F}z = (\omega_0(z), \omega_1(z), \dots, \omega_{n-1}(z)).$$

- (a) Show that $\mathcal{F}$ is a unitary operator on $\mathbb{C}^n$.
- (b) Show that if $(z_0, \dots, z_{n-1}) \in \mathbb{C}^n$ and z_n is defined to equal z_0 , then

$$\mathcal{F}^{-1}(z_0, z_1, \dots, z_{n-1}) = \mathcal{F}(z_n, z_{n-1}, \dots, z_1).$$

- (c) Show that $\mathcal{F}^4 = I$.

Axler Note. The discrete Fourier transform has many important applications in data analysis. The usual Fourier transform involves expressions of the form $\int_{-\infty}^{\infty} f(x) e^{-2\pi i t x} dx$ for complex-valued integrable functions f defined on $\mathbb{R}$.

Solution. Let $e_0, e_1, \dots, e_{n-1}$ be the standard orthonormal basis of $\mathbb{C}^n$, so the j th component of $\mathcal{F}e_k$ is

$$\omega_j(e_k) = \frac{1}{\sqrt{n}} \sum_{p=0}^{n-1} (e_k)_p e^{-2\pi i j p / n} = \frac{1}{\sqrt{n}} e^{-2\pi i j k / n},$$

where $(e_k)_p = \delta_{pk}$.

(a) Let $m, k \in \{0, 1, \dots, n-1\}$. Then

$$\langle \mathcal{F}e_m, \mathcal{F}e_k \rangle = \sum_{j=0}^{n-1} \omega_j(e_m) \overline{\omega_j(e_k)} = \sum_{j=0}^{n-1} \frac{1}{\sqrt{n}} e^{-2\pi i j m / n} \cdot \frac{1}{\sqrt{n}} e^{2\pi i j k / n} = \frac{1}{n} \sum_{j=0}^{n-1} e^{2\pi i j (k-m) / n}.$$

If $m = k$, then $\langle \mathcal{F}e_m, \mathcal{F}e_k \rangle = 1$. Otherwise, let $\zeta = e^{2\pi i (k-m) / n}$. Then $\zeta^n = 1$ and $\zeta \neq 1$, since $0 < |k-m| < n$. Therefore,

$$\langle \mathcal{F}e_m, \mathcal{F}e_k \rangle = \frac{1}{n} \sum_{j=0}^{n-1} \zeta^j = \frac{1}{n} \cdot \frac{1 - \zeta^n}{1 - \zeta} = 0.$$

Hence, $\mathcal{F}e_0, \dots, \mathcal{F}e_{n-1}$ is an orthonormal list in $\mathbb{C}^n$. By 6.28, $\mathcal{F}e_0, \dots, \mathcal{F}e_{n-1}$ is an orthonormal basis of $\mathbb{C}^n$, so $\mathcal{F}$ is unitary on $\mathbb{C}^n$ by 7.53(d).

(b) Define $R \in \mathcal{L}(\mathbb{C}^n)$ by $R(z_0, z_1, \dots, z_{n-1}) = (z_{n-1}, z_{n-2}, \dots, z_0)$. Then for any $z \in \mathbb{C}^n$ and $j \in \{0, \dots, n-1\}$,

$$(\mathcal{F}^2 z)_j = \frac{1}{\sqrt{n}} \sum_{m=0}^{n-1} (\mathcal{F}z)_m e^{-2\pi i j m / n} = \frac{1}{n} \sum_{m=0}^{n-1} \sum_{k=0}^{n-1} z_k e^{-2\pi i m (k+j) / n} = \sum_{k=0}^{n-1} z_k \left(\frac{1}{n} \sum_{m=0}^{n-1} e^{-2\pi i m (k+j) / n} \right)$$

where $r_k = e^{-2\pi i (k+j) / n}$. Observe that $0 \leq k+j \leq 2n-2$, so $r_k \neq 1$ unless $k+j=0$ or $k+j=n$ in which case the inner sum evaluates to 1. Otherwise, we have $r_k \neq 1$ and $r_k^n = 1$, so

$$\frac{1}{n} \sum_{m=0}^{n-1} r_k^m = \frac{1}{n} \cdot \frac{1 - r_k^n}{1 - r_k} = 0.$$

Observe that for $k \in \{0, \dots, n-1\}$, one such k exists for which $k+j=0$ or $k+j=n$, namely $k=0$ if $j=0$ or $k=n-j$ if $1 \leq j \leq n-1$. Hence, $(\mathcal{F}^2 z)_0 = z_0$ and for $j \geq 1$, $(\mathcal{F}^2 z)_j = z_{n-j}$, whereupon $\mathcal{F}^2 z = Rz$. Since R fixes the 0th coordinate and reverses the j th and $(n-j)$ th coordinates for $j \geq 1$, R is its own inverse, i.e., $R^2 = I$. From this, we get $\mathcal{F}(\mathcal{F}R) = \mathcal{F}^2 R = R^2 = I$. By 3.68, $(\mathcal{F}R)\mathcal{F} = I$, and hence $\mathcal{F}^{-1} = \mathcal{F}R$. Because $z_n = z_0$,

$$\mathcal{F}^{-1}(z_0, z_1, \dots, z_{n-1}) = (\mathcal{F}R)(z_0, z_1, \dots, z_{n-1}) = \mathcal{F}(z_0, z_{n-1}, \dots, z_1).$$

(c) By part (b), $\mathcal{F}^2 = R$ and $R^2 = I$, so $\mathcal{F}^4 = R^2 = I$. ■

Exercise 7D20

Suppose A is a square matrix with linearly independent columns. Prove that there exist unique matrices R and Q such that R is lower triangular with only positive numbers on its diagonal, Q is unitary, and $A = RQ$.

Solution. Since A is n -by- n and has linearly independent columns, $\text{column rank } A = n$. By 3.57, $\text{row rank } A = \text{column rank } A = n$, so the rows of A are linearly independent as well. Since the columns of A^* are the conjugate transposes of the rows of A , the columns of A^* are linearly independent. By 7.58, there exist unique matrices Q_1 and R_1 such that Q_1 is unitary, R_1 is upper triangular with positive diagonal entries, and $A^* = Q_1 R_1$. Taking conjugate transposes gives $A = R_1^* Q_1^*$, where R_1^* is lower

triangular with the same positive diagonal entries as R_1 , and Q_1^* is unitary by 7.57(d). Let $R = R_1^*$ and $Q = Q_1^*$, so $A = RQ$ where R and Q have the desired properties.

To see uniqueness of R and Q , suppose $A = RQ = R_0Q_0$ where R_0 is lower triangular with positive diagonal entries and Q_0 is unitary. Taking conjugate transposes again yields $A^* = Q^*R^* = Q_0^*R_0^*$, where R^* and R_0^* are upper triangular with positive diagonal entries and Q^* and Q_0^* are unitary. By the uniqueness part of 7.58, we must have $R^* = R_0^*$ and $Q^* = Q_0^*$, which implies $R = R_0$ and $Q = Q_0$. ■